WORKED SOLUTION

# STEP II 2007, Question 14

Mechanics • Pdf

**STEP II 2007, Question 14**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q14

## The key idea

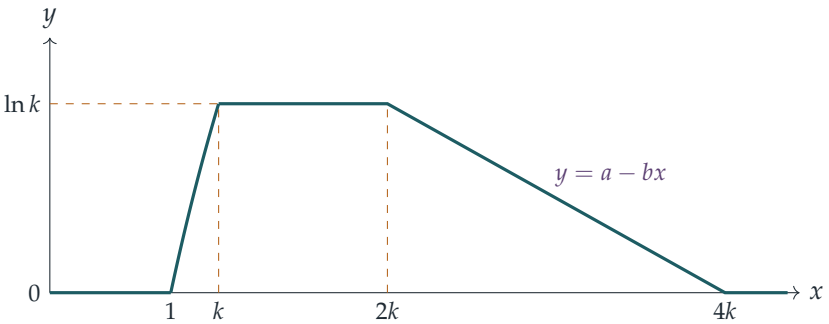
The word *continuous* is doing a lot of work. A density defined in pieces has to join up at every break point, and that gives us two equations for $a$ and $b$ straight away. The one remaining condition, that the total area under the graph is 1, then pins down $k$. For the median we add up areas piece by piece until we pass one half.

## Solution

Throughout, $k > 1$: otherwise the interval $1 \leq x \leq k$ would be empty, and in any case we need $\ln k > 0$ for the density to be non-negative and not identically zero beyond $x = 1$.

### Part (i)

On $[1, k]$ the graph is the usual logarithm curve, starting at 0 when $x = 1$ (so it joins the zero piece continuously), increasing and concave. It reaches height $\ln k$ at $x = k$, stays level at $\ln k$ until $x = 2k$, then falls in a straight line to 0 at $x = 4k$ (we show below that continuity forces exactly this), and is zero afterwards. The sketch uses the values we find in part (ii): $k = \mathrm{e}^{1/3} \approx 1.40$, so the plateau is at height $\frac{1}{3}$.



**Part (ii)**

*Continuity.* At $x = k$ both formulas give $\ln k$, so nothing new there. At $x = 2k$ and $x = 4k$ continuity requires

$$a - 2bk = \ln k, \qquad a - 4bk = 0.$$

Subtracting, $2bk = \ln k$, so

$$b = \frac{\ln k}{2k}, \qquad a = 4bk = 2\ln k.$$

*Total probability.* We split the area into three pieces. Integrating by parts (with $u = \ln x$, $\mathrm{d}v = \mathrm{d}x$),

$$\int_1^k \ln x \,\mathrm{d}x = \Big[x\ln x - x\Big]_1^k = k\ln k - k + 1.$$

The plateau is a rectangle of width $k$ and height $\ln k$, with area $k\ln k$. The last piece is a triangle with base $4k - 2k = 2k$ and height $\ln k$, with area $\frac{1}{2}\cdot 2k \cdot \ln k = k\ln k$. So

$$1 = (k\ln k - k + 1) + k\ln k + k\ln k = 3k\ln k - k + 1,$$

which simplifies to

$$k(3\ln k - 1) = 0.$$

Since $k > 1$, we cannot have $k = 0$, so $\ln k = \frac{1}{3}$ and

$$k = \mathrm{e}^{1/3} \approx 1.396, \qquad a = 2\ln k = \tfrac{2}{3}, \qquad b = \frac{\ln k}{2k} = \frac{1}{6\,\mathrm{e}^{1/3}} \approx 0.119.$$

Note $b > 0$, so the last piece really does slope downwards, as drawn.

**Part (iii)**

We find which piece contains the median $m$ by accumulating probability. Using $\ln k = \frac{1}{3}$,

$$\mathrm{P}(X \le k) = k\ln k - k + 1 = 1 - \tfrac{2k}{3} \approx 0.070,$$

and the plateau adds $k\ln k = \frac{k}{3}$, so

$$\mathrm{P}(X \le 2k) = 1 - \tfrac{2k}{3} + \tfrac{k}{3} = 1 - \tfrac{k}{3} \approx 0.535.$$

? **Ask yourself** Which piece of the graph does the median lie on?

Since $0.070 < \frac{1}{2} < 0.535$, the median lies on the plateau, $k \le m \le 2k$. There the density is constant at $\frac{1}{3}$, so

$$\mathrm{P}(X \le m) = 1 - \tfrac{2k}{3} + \tfrac{1}{3}(m - k) = \tfrac{1}{2}.$$

Multiplying by 3: $3 - 2k + m - k = \frac{3}{2}$, so

$$m = 3k - \tfrac{3}{2} = 3\mathrm{e}^{1/3} - \tfrac{3}{2} \approx 2.69.$$

This does lie between $k \approx 1.40$ and $2k \approx 2.79$, as it must.

## Checks

- Numerically integrating the density with $k = \mathrm{e}^{1/3}$, $a = \frac{2}{3}$, $b = 1/(6\mathrm{e}^{1/3})$ (midpoint rule) gives total area 1.0000000, and $\int_1^{2.68684} f(x)\,\mathrm{d}x = 0.5000000$.
- Sanity check on the median: just over half the probability lies to the left of $2k$ (about 0.535), so the median should sit a little to the left of $2k \approx 2.79$; indeed $m \approx 2.69$.

## Common slips

- Forgetting the lower limit in $\int_1^k \ln x\,\mathrm{d}x$: the $+1$ from $-[x\ln x - x]$ at $x = 1$ is what makes the equation for $k$ factorise so neatly.
- Assuming the median is on the sloping piece because it is "in the middle" of the support $[1, 4k]$; always accumulate the areas first.

## Beyond STEP

Here the density was built from pieces and the constants were fixed by continuity and normalisation. At university you will meet the *cumulative distribution function* $F(x) = \mathrm{P}(X \leq x)$ as the basic object: it exists for every random variable, discrete or continuous, and quantities like the median are simply solutions of $F(m) = \frac{1}{2}$. Grimmett and Welsh, *Probability: An Introduction*, develops this viewpoint clearly.

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